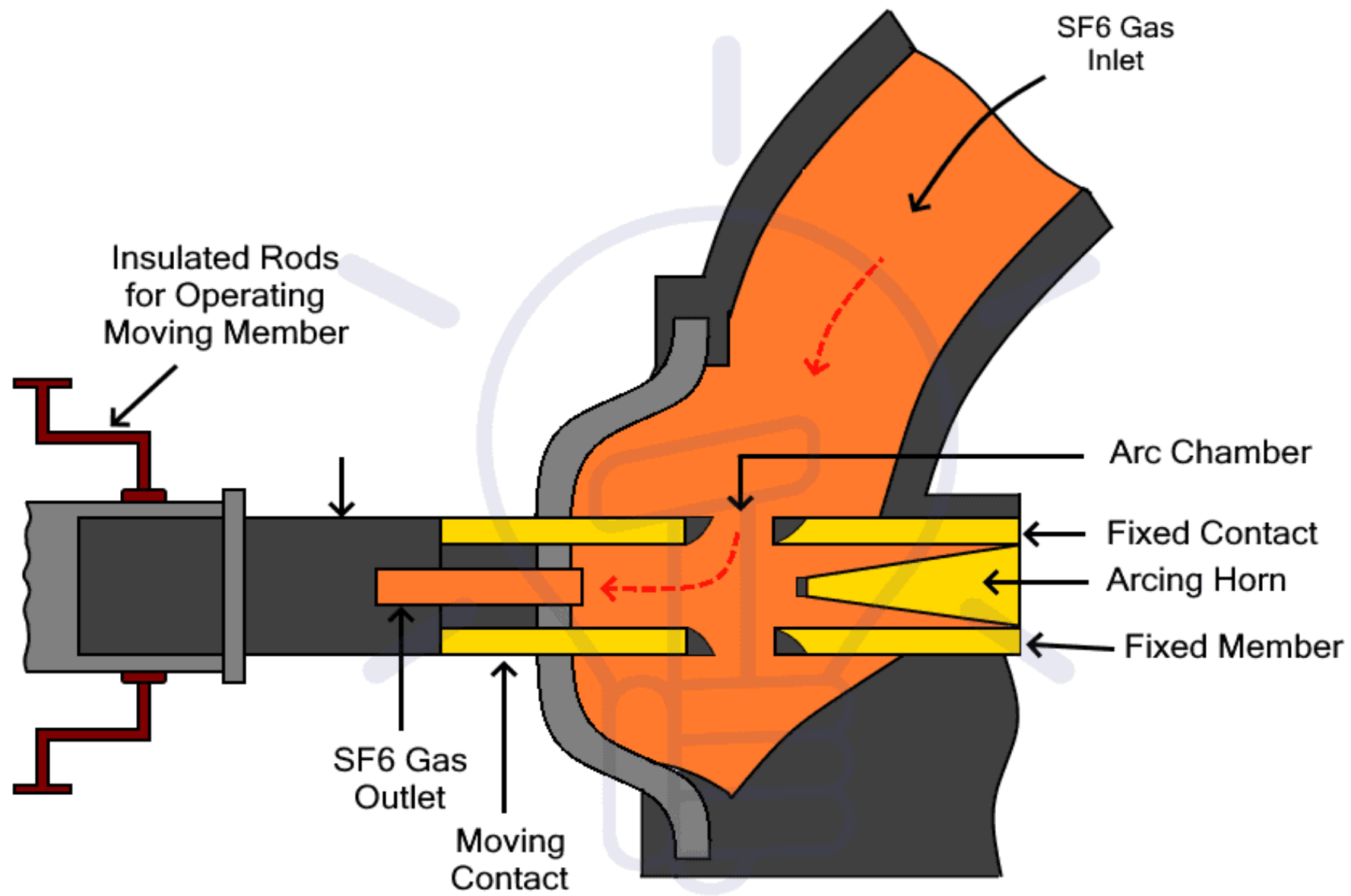


2.2. Sulphur Hexafluoride (SF6) Circuit Breaker

Sulphur Hexafluoride or shortly known as SF6 is a non-flammable and insulating gas that has very high electronegativity. It has a high tendency to absorb electrons.

When the arc is struck between the contacts, the medium is ionized due to free electrons. The SF6 absorbs the free electrons and forms negative ions that are far heavier than free electrons. Due to their heavyweight, they are static and reduces the flexibility of charges. This improves the dielectric strength of the medium where the arc is quenched. The SF6 has far more superior insulating and arc quenching capabilities than air, almost 100 times better.



Sulphur Hexafluoride SF6 Circuit Breaker

The SF6 is very expensive and a greenhouse gas. SF6 itself is not poisonous but its product gases are poisonous and Its emission is dangerous for the environment. Therefore, a close circuit gas system is designed for such breaker where the SF6 gas is re-utilized after each operation. The system also monitors its pressure, which is directly proportional to its dielectric strength.

There are three types of **SF6 Circuit Breakers**

2.2.1 Non-Puffer Type SF6 Circuit Breaker

2.2.2 Single Pressure Puffer Type SF6 Circuit Breaker

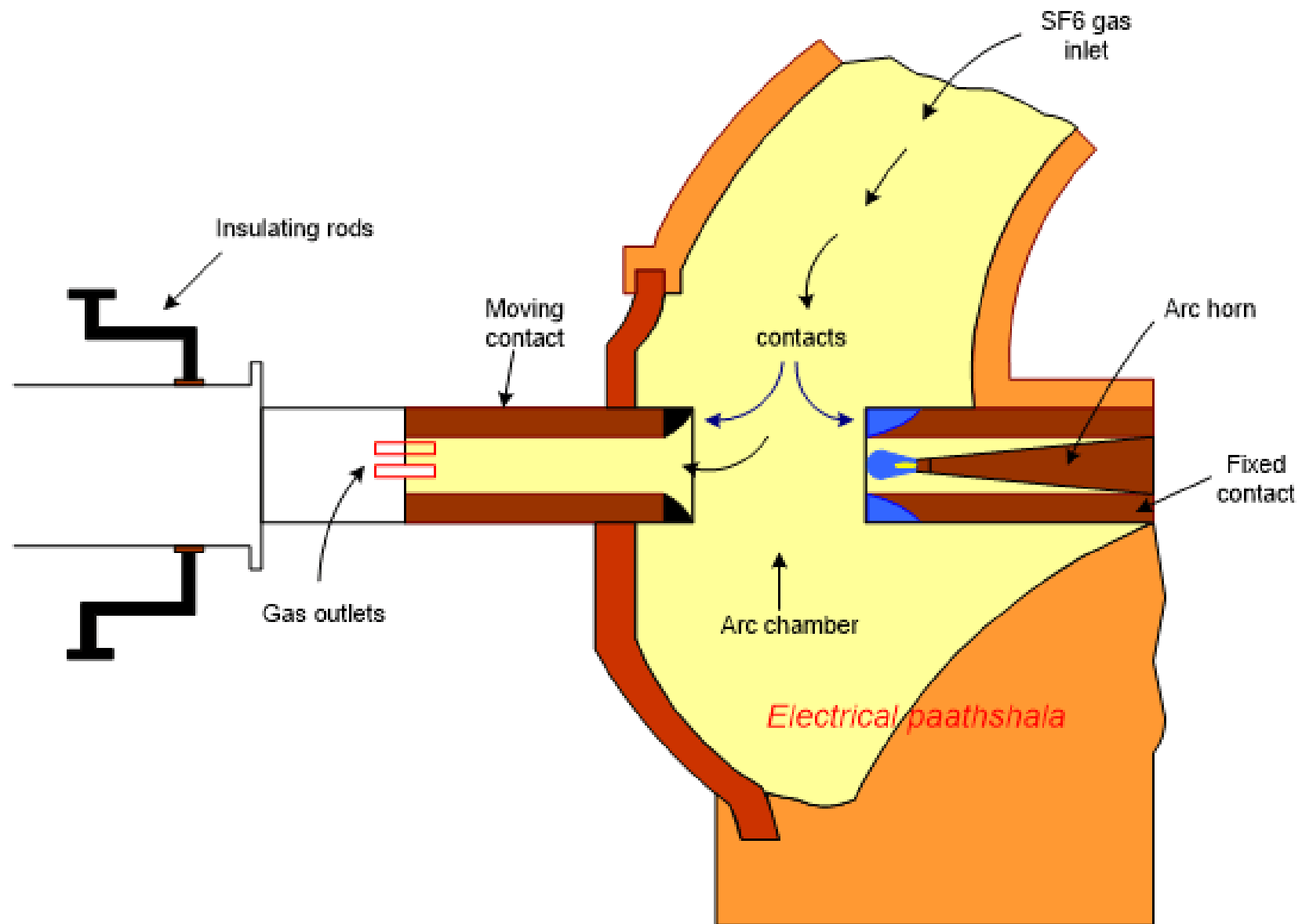
2.2.3 Double Pressure Type SF6 Circuit Breaker

2.2.1 Non-Puffer Type

In non-Puffer SF6 CB, pressurized SF6 gas is stored in the gas chamber.

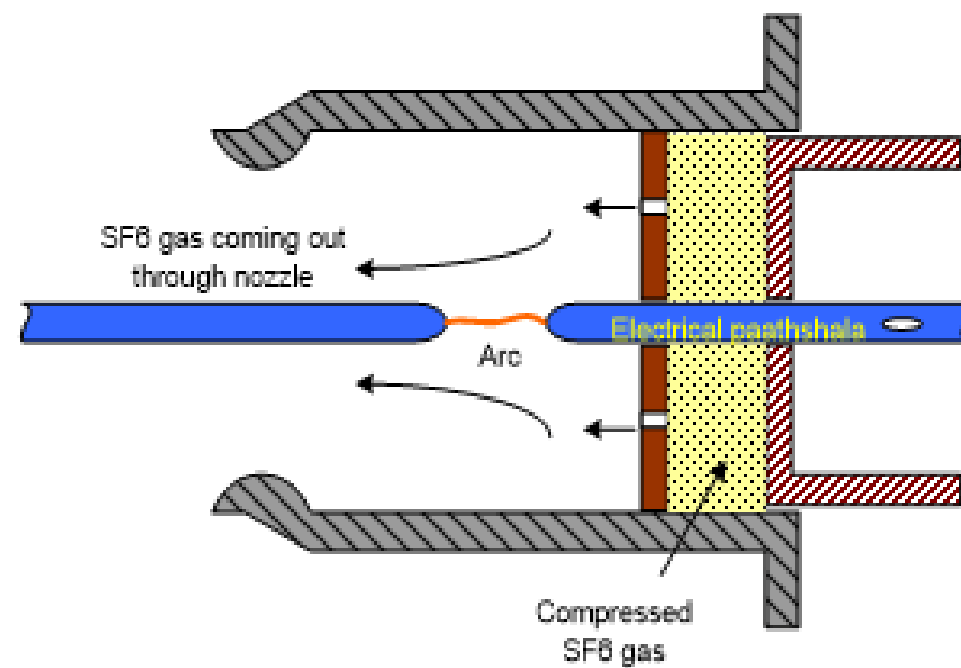
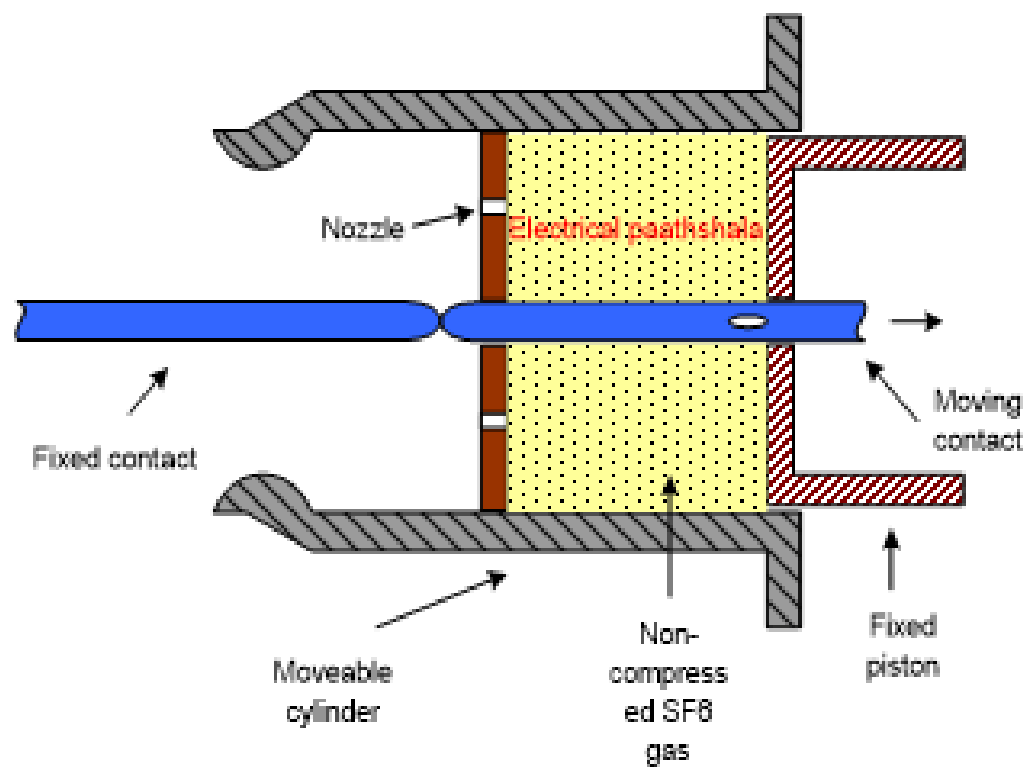
At the same time, the arc quenching is done inside an interrupter unit. This unit has moving and fixed contacts that are basically hollow cylinders. The fixed contact has arc horns, while the moving contact has vents for pressurized gas to flow.

Under a fault condition, the moving contact is moved apart from the fixed contact. Its movement is synchronized with the valve of the gas chamber. As soon as the contacts open, the valve is opened, and pressurized SF6 is introduced into the arc chamber. The SF6 quenches the arc and flows through the hollow moving contact. This gas is then recombined and pumped back into the gas chamber for re-utilization.



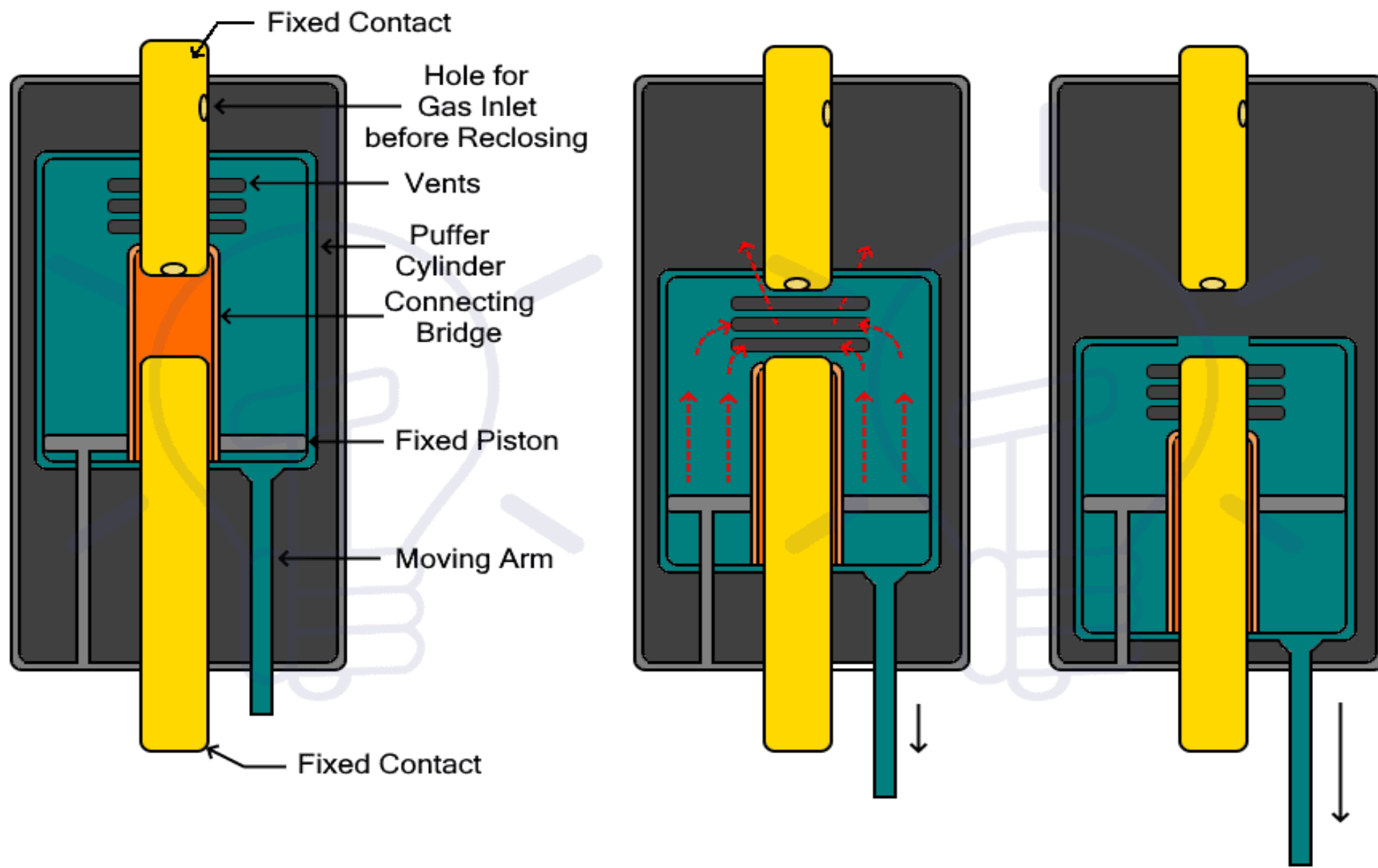
2.2.2 Single Pressure Puffer Type SF6 Circuit Breaker

Such type of SF6 CB has a Puffer cylinder. A puffer cylinder is a hollow cylinder that acts as a bridge between two fixed contacts. The cylinder can slide upward and downward axially along the contacts. There is a stationary piston inside the puffer cylinder as shown in the figure. By moving the cylinder, its internal volume can be varied. The movable cylinder is filled with SF6 gas. During circuit breaking operation, the cylinder moves downward against the piston to break the connection between the fixed contacts. Consequently, it reduces the volume inside the cylinder, compressing the SF6 inside it.



At this point, the upper fixed contact blocks the vents in the puffer cylinder, therefore the SF6 cannot flow out. By moving the cylinder further down, the vents are unblocked and the contacts are opened. An arc is generated as well as the SF6 starts flowing out through the hollow contacts. This pressurized SF6 quenches the arc.

The same process is done in reverse to close the circuit. by moving the cylinder upwards, its volume is increased as well as a low pressure is generated. Due to this, the SF6 from the surrounding flows into the cylinder through the contacts and the vents.

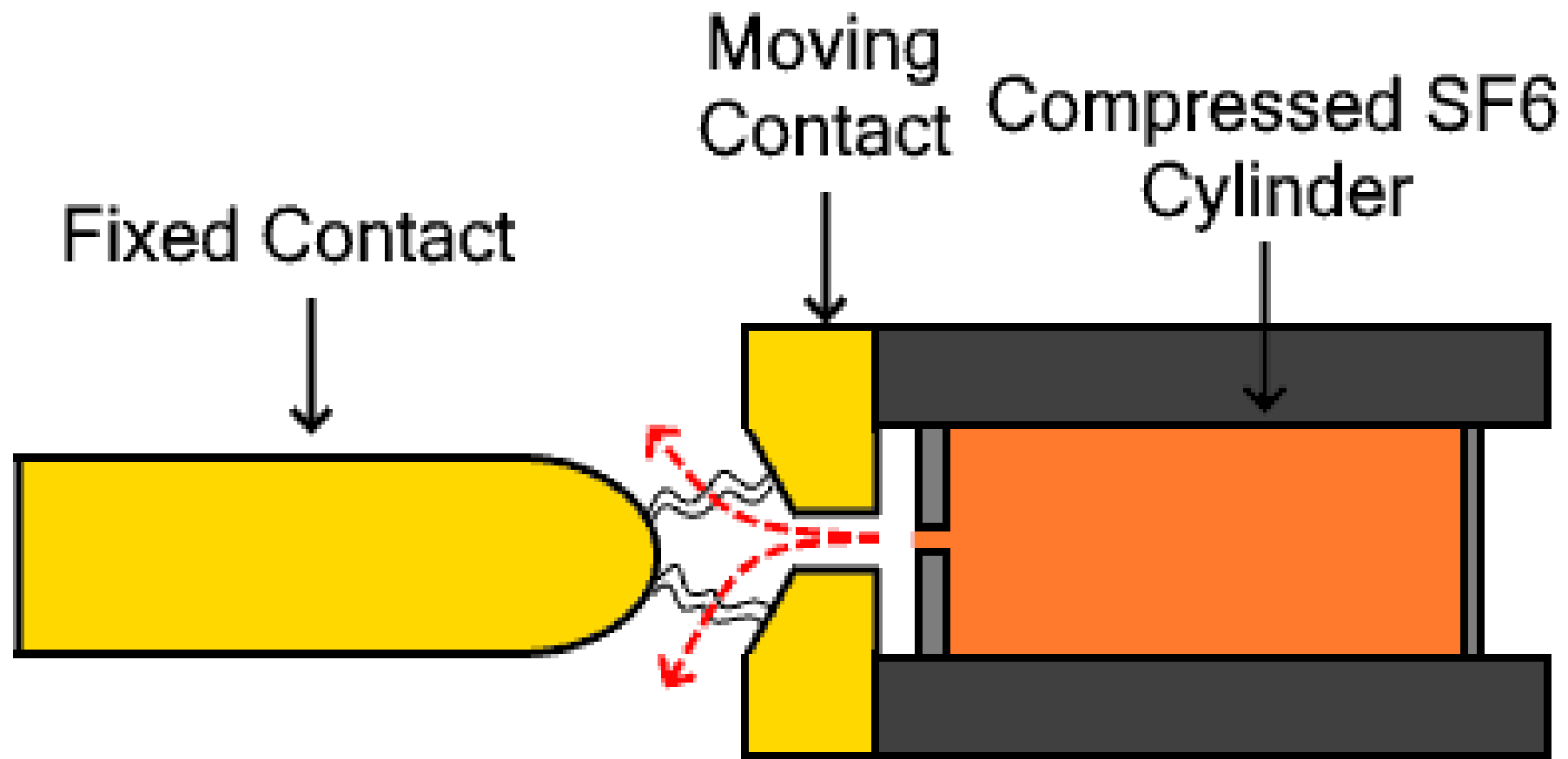


Puffer type SF6 Circuit Breaker

2.2.3 Double Pressure Type SF6 Circuit Breaker

Such types of SF6 Circuit breakers are **obsolete**. In such breakers, the gas is compressed and stored in a chamber, which is released upon opening of the contacts to quench the arc. **It has the same operation as an air blast CB, except the gas is recompressed and stored in the gas chamber.**

There is an SF6 cylinder inside the moving contact that is blocked by fixed contact in the closed position. When the contacts separate, an arc is produced. **At the same time, the highly pressurized gas from the cylinder rushes out to the low-pressure area.** The blast of SF6 quenches the arc as discussed earlier. **The gas is filtered, compressed, and restored in the cylinder for reuse. It is obsolete due to the complicated gas system required to maintain the gas. It also includes a heater in case the gas is liquefied due to low temperature.**



Double Pressure SF6 Circuit Breaker

Advantages of SF6 Circuit Breaker

- Sulphur Hexafluoride SF6 has superior arc quenching property almost 100 times more effective than air.
- The arcing time of the SF6 circuit breaker is very short.
- The dielectric strength of SF6 gas is 2 to 3 times higher than air. It also increases with increasing pressure.
- Due to high dielectric strength, the required contact separation is small to prevent arc restriking.
- High dielectric strength leads to large current interruption capabilities.
- The SF6 CB has a compact design. Thus require small space and cost for installation.

- SF6 gas can handle all kinds of switching phenomena.
- SF6 CB has a closed circuit gas system with no leakage. Therefore, best for any installation in any kind of (extreme) environment.
- No carbon particles are formed with arcing, thus the dielectric strength does not reduce.
- It does not require an expensive and bulky air compressing system except for double pressure type which is obsolete.
- The operation of SF6 CB is noiseless.
- SF6 gas is non-toxic in its pure state.
- SF6 gas is non-flammable, thus no chance of fire hazards.
- Since its operation is flawless, it requires less maintenance.

Disadvantages of SF6 Circuit Breaker

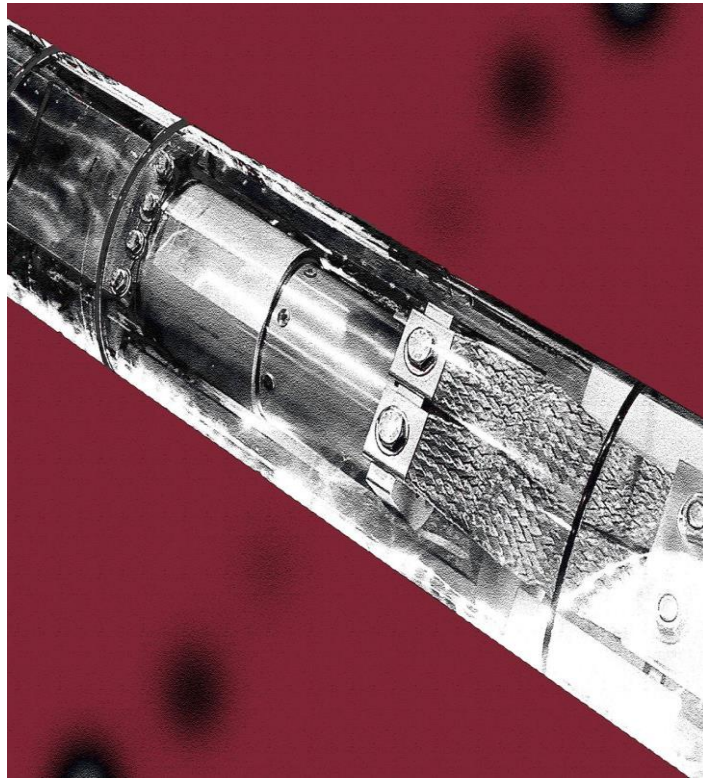
- The byproducts formed from SF6 gas during arcing are toxic for the environment but they are mostly recombined into SF6
- The decomposed SF6 is toxic.
- SF6 is an expensive gas so these circuit breakers are costly.
- Leakage of SF6 from the joints must be continuously monitored.
- It requires special transportation and maintaining of the quality of gas.
- The SF6 is heavier than oxygen and can cause difficulty in breathing.
- Recombination and reconditioning of the SF6 gas requires additional equipment.

2.3. Carbon dioxide (CO_2) circuit breaker

A carbon dioxide (CO_2) circuit breaker is a high-voltage electrical switch that uses a mixture of gases, often including CO_2 , to extinguish the electrical arc that forms when the circuit is opened. These breakers are being developed as an environmentally friendly alternative to traditional ones that use [sulfur hexafluoride \(\$SF_6\$ \)](#), a potent greenhouse gas. Research is focused on using CO_2 in various phases, including supercritical CO_2 , to create a breaker with good arc quenching capabilities and a lower carbon footprint.

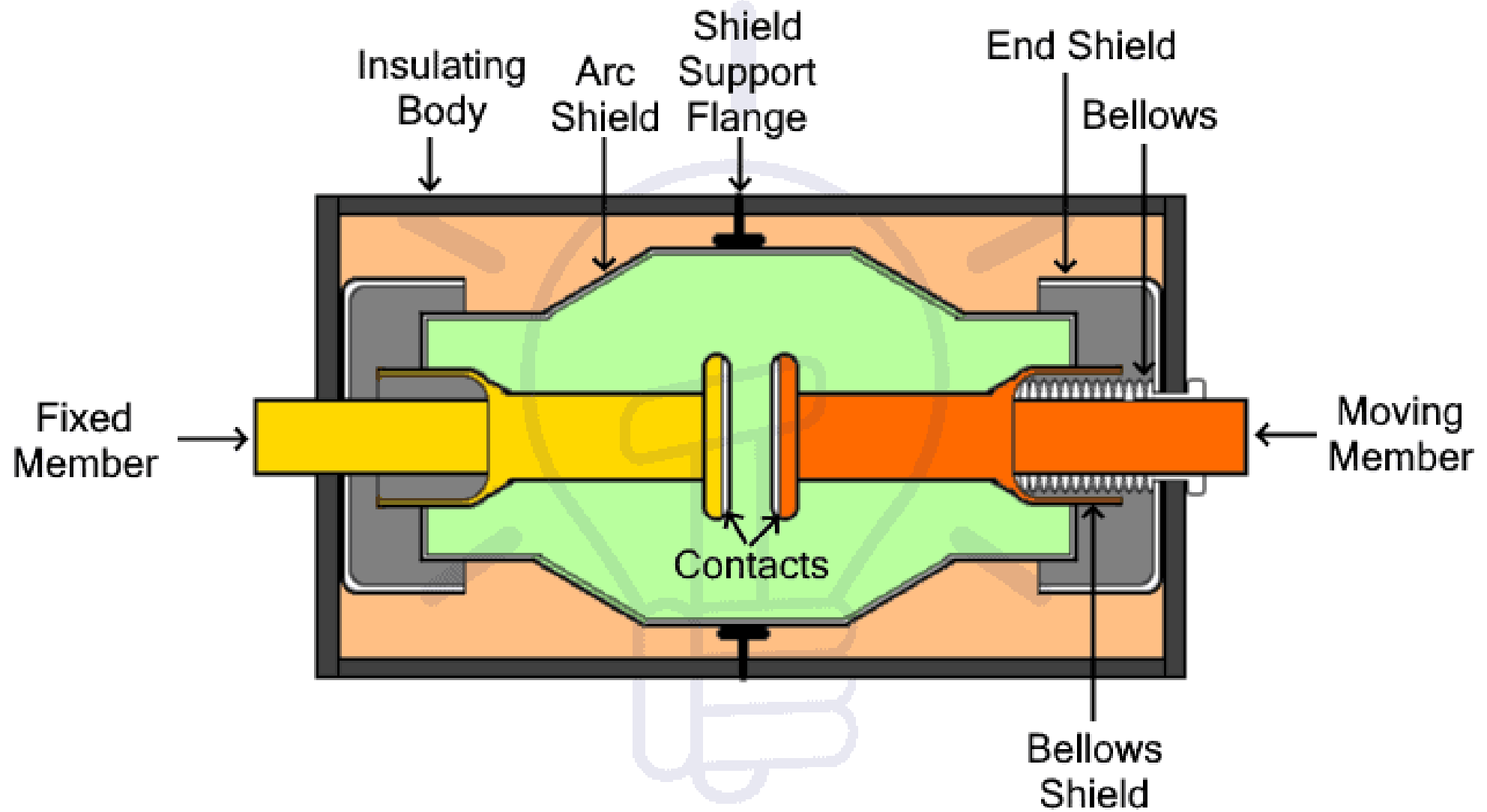
These devices are placed throughout substations, power-generation plants, [transmission and distribution](#) networks, and industrial facilities where equipment operates **in tens to hundreds of kilovolts**.

A high-voltage circuit breaker interrupter, stops current by mechanically creating a gap and an arc, and then blasting high-pressure gas through the gap. This halts the current by absorbing free [electrons](#) and quenching the arc as the [dielectric](#) strength of the gas is increased.



2.4. Vacuum Circuit Breaker (VCB)

A Vacuum Circuit Breaker or VCB is a type of circuit breaker that uses the vacuum as the arc quenching medium. The vacuum has very high dielectric strength and arc quenching properties far more superior than any other medium. It can quickly recover its dielectric strength. Due to its high dielectric strength, VCB requires a very small gap between its contacts to prevent re-striking. It is suitable for switching medium-voltage ranging from 22kV to 66 kV.



Vacuum Circuit Breaker

The **switching operation** of current-carrying contacts and the arc interruption occurs inside a closed chamber called **Vacuum interrupter**. Its outer insulating body is made up of glass or ceramic material. It consists of fixed and movable contacts surrounded by arc shield.

The arc shield is used for the prevention of the deterioration of the dielectric strength of the vacuum by preventing the ionized metallic vapors on the inner side of the outer insulating body. The movable member is connected to a controlled mechanism (for movement) using bellow. The bellow completely seals the vacuum chamber and prevents any leak.

The **operation** of VCB is very simple, the arc interruption occurs inside the vacuum upon first zero current. When a fault condition occurs, the contacts are separated. During separation, the contacts do not separate at once but its contact area is reduced which eventually reduces to a single point.

The amount of current passing through this single point heats up the contact and vaporizes (reducing the dielectric strength of the vacuum) to create a medium for the arc. Thus the arc is generated. At next zero current, the conducting metallic vapors re-condenses on the contact surface and the dielectric strength of the vacuum is recovered.

Since the contacts are separated and there are no vapors between them, the arc cannot re-strike. In simple words, a VCB quenches the arc by producing the high dielectric strength to prevent arc re-striking after the current zero.

Since the arc is generated due to the ionization of the contacts, its material plays a vital role in keeping the CB reliable and maintenance-free. Therefore, the VCB contact material must have the following properties

- It must have high electrical conductivity to avoid overheating upon normal load currents.
- It must have high thermal conductivity to dissipate large heat produced during arcing.
- It should have high arc withstanding capabilities and low current chopping level.
- It must have low resistance with high density.

The contacts are made from copper alloy such as Copper-bismuth, copper-lead, and copper-chromium material.

Advantages of VCB

- Vacuum is the absence of matter thus VCB has no fire hazards
- The Vacuum has very high dielectric strength and superior arc quenching properties than air and SF6.
- Due to high dielectric strength, VCB requires a small contact gap to extinguish the arc.
- VCB is compact and requires small installation space.
- It is free of maintenance thus it's reliable having a long life.
- It does not produce any noise doing the operation.
- There are no toxic exhaust gases.
- Its operation is very fast.
- It is suitable for repeated use.

- . It can break all types of fault currents.
- . Since there is a vacuum, the control mechanism requires less [power](#) to move the contacts.

Disadvantages of VCB

- A single vacuum interrupter can only interrupt up to 38kV.
- For interrupting voltages more than 38kV, multiple vacuum interrupters must be connected in series.
- It's uneconomical for voltages exceeding 38Kv since it requires more than one VCB and the total cost increases.
- In case of loss of vacuum, the VCB becomes useless.